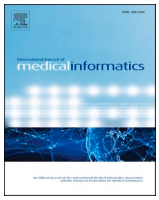
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Review article



Machine learning to predict stroke risk from routine hospital data: A systematic review

William Heseltine-Carp [a,*], Megan Courtman [a,b], Daniel Browning [a], Aishwarya Kasabe [a], Michael Allen [c], Adam Streeter [d], Emmanuel Ifeachor [d,e], Martin James [f], Stephen Mullin [a]

[a] University of Plymouth, Room N6, ITTC Building, Plymouth Science Park, Plymouth PL68BX, UK
[b] University of Plymouth, Plymouth PL4 8AA, UK
[c] University of Exeter, Medical School, St Lukes Campus, Heavitree Road, SC 2.30, Exeter EX4 4QJ, UK
[d] University of Plymouth, N15, ITTC1, Plymouth Science Park, Plymouth PL6 8BX, UK
[e] School of Engineering, Computing and Mathematics, University of Plymouth, Plymouth PL4 8AA, UK
[f] University of Exeter, Academic Department of Healthcare for Older People, Royal Devon & Exeter Hospital, Exeter EX2 5DW, UK

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ABSTRACT

Purpose: Stroke remains a leading cause of morbidity and mortality. Despite this, current risk stratification tools such as CHA_2DS_2-VASc and QRISK3 are of limited accuracy, particularly in those without a diagnosis of atrial-fibrillation. Hence, there is a need for more accurate stroke risk prediction models. Machine-learning (ML) may provide a solution to this by leveraging existing routine hospital databases to build accurate stroke risk prediction models and identify novel risk factors for stroke.

Aims: In this systematic review we appraise current research using ML to predict stroke risk from routine hospital data. Based on these findings we then highlight common methodological limitations and recommendations for future research.

Methods: In this review we identify 49 original research (38 in the general population and 11 in AF specific populations) articles from the PUBMED database from January-2013 to December-2024 using ML and routine hospital data to predict the risk of stroke.

Results: ML models were able to accurately predict stroke risk in both AF specific and general populations, with AUCs ranging from 0.64 to 0.99. Where tested, ML also consistently outperformed traditional risk stratification tool, such as CHA_2DS_2-VASc. ML also appeared useful in identifying several novel risk factors from electrocardiogram, laboratory test and echocardiography data.

However, the quality of datasets were often limited, there was a high suspicion of overfitting and models often lacked calibration, external validation and explainability analysis.

Conclusion: Whilst ML has shown great potential in stroke prediction and identifying novel risk factors for stroke, improvements in study methodology is required prior to integration of ML into routine healthcare. Future research should adhere to the EQUATOR guidance on prediction models and encourage interdisciplinary collaboration between computer scientists and clinicians. Further prospective RCTs are also required to validate models in the clinical setting and the identify barriers of integrating ML into routine healthcare.

1. Background

1.1. Current stratification tools for stroke risk.

Stroke represents the second leading cause of death worldwide, affecting approximately 15 million individuals per year [1]. Several risk factor scores are currently available to clinicians to stratify stroke risk and guiding clinical decision making. These include, the Cardiovascular-Risk-Score-3 (QRISK3), CHA_2DS_2-VASc and until recently, the ABCD2 score which predicts stroke risk following a transient ischaemic attack

* Corresponding author at: Room N6, ITTC building, Plymouth Science Park, Plymouth PL68BX, UK.
E-mail addresses: w.heseltine-carp@nhs.net (W. Heseltine-Carp), megan.courtman@plymouth.ac.uk (M. Courtman), daniel.browning@nhs.net (D. Browning), aishwarya.kasabe@plymouth.ac.uk (A. Kasabe), m.allen@exeter.ac.uk (M. Allen), adam.streeter@plymouth.ac.uk (A. Streeter), e.ifeachor@plymouth.ac.uk (E. Ifeachor), martinjames@nhs.net (M. James), stephen.mullin@plymouth.ac.uk (S. Mullin).

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(TIA) [2–4]. The receiver-operator-characteristic-area-under-curve (ROC-AUC) of these tools range from 0.67 to 0.76 and have proven useful in the clinical setting, however 25–30 % of people who suffer stroke have no known risk factors [5]. Indeed, there is a need for improvement of these risk scores. CHA_2DS_2-VASc has been criticised for its poor specificity [6], which is an important consideration since the use of anticoagulants carries a significant risk of bleeding [7]. The QRISK3 score has been criticised for underestimating cardiovascular risk and the ABCD2 score has demonstrated poor sensitivity and specificity, hence it is no longer recommended in UK guidelines by National-Institute-of-Healthcare-Excellence (NICE) [8].

These limitations can be attributed to a number of factors. Firstly, most of these models are not generalisable to the average population and are instead targeted for use in those with a history of atrial fibrillation (AF) or transient-ischaemic-attack (TIA) [6]. Secondly, their scoring tools exclude several risk factors for stroke, such as smoking, obesity and a family history of stroke. These variables were excluded due to their relative lack of significant effect during construction of these instruments [9]. However, these variables remain clinically relevant and collectively explain a significant proportion of stroke risk [10]. Also, their reliance on established epidemiological risk factors may exclude novel stroke risk factors which may be found by analysis of other data sources, such as imaging and laboratory tests. Thirdly, dichotomization of risk variables to present or absent (rather than a continuous spectrum) and not including interactions between variables, sacrifices the predictive power of models [11]. Finally, these models aim to predict the risk of only ischaemic stroke. Despite the existence of known risk factors for haemorrhagic stroke, no instrument currently predicts the risk of haemorrhagic stroke [12]. These issues highlight the unmet need for the development of models that are able to accurately predict the risk of both haemorrhagic and ischaemic stroke amongst patients with and without a history of AF, TIA and/or cardiovascular disease. One such field that shows promise in the development of such models is artificial-intelligence (AI).

1.2. Artificial-intelligence

AI involves the process of perceiving, processing, and inferring information by machines or computer programs [13]. In the last decade we have begun to see the capabilities of AI in impacting and improving patient care, both in the context of stroke and other disease areas. Examples include e-ASPECTS (Brainomix limited), which is able to successfully identify possible candidates for thrombolysis and/or thrombectomy [14] by interpreting computerised-tomography (CT) brain scans and “InnerEye (Microsoft)” which processes radiological investigations, identifying likely cancers and creating radiotherapy plans for patients [15].

Today, large and varied datasets of routinely collected medical data are readily available, which act as an attractive substrate for understanding the unexplained factors and/or interaction of factors driving stroke risk.

Within these large datasets exists a wealth of variables. These variables are able to influence a given outcome directly, or indirectly as a result of complex interactions between one-another. In some cases, AI is capable of analysing these complex interactions and gaining insight into how variables within a dataset interact and affect the overall outcome [16]. This enables identification of both major and minor risk factors for stroke and an understanding of how these factors interact with each other to influence overall risk.

AI may therefore be invaluable in building a comprehensive tool to predict stroke risk and guide clinical decision making.

Of the various categories of AI that exist, machine learning (ML) is the most commonly used in predictive modelling within the healthcare setting. ML is a subcategory of AI in which algorithms learn to improve their performance through exposure to data alone [16]. **Supervised learning (SL)** is the branch of ML which is most relevant to predictive modelling. SL uses labelled datasets to train algorithms to accurately classify data and predict outcomes. It is the technology with most applicability to predictive modelling of stroke [16]. Several SL methods exist. Table 1 lists the most commonly used SL methods for predicting the risk of stroke from routine hospital data.

Because many SL models are capable of providing insight into variable importance, certain SL models are well suited to disease prediction and prognosis, novel risk factor identification, drug discovery, medical image analysis and disease diagnosis. They are hence often the models of choice for predicting the risk of stroke, however they require accurately labelled and organised datasets, and have been criticised for a lack of generalisability within medical research [17]. In this paper we provide an overview of the current research using supervised ML to predict stroke risk and identify novel risk factors for stroke.

1.3. Aims

Our aims are outlined as follows:

1. To evaluate the effectiveness of ML in predicting stroke risk and identifying novel risk factors for stroke in patients with and without AF.
2. To identify and suggest areas of methodological improvement when building ML models to predict stroke risk.
3. To identify areas of future study in using ML to predict stroke risk.

2. Materials and methods

This systematic review was conducted in accordance with guidelines outlined by Preferred-Reporting-Items-for-Systematic-Reviews-and-Meta-Analyses (PRISMA) [18].

Data search

2.1. Search protocol

A PUBMED search was used to identify relevant full-text research articles. The following search input was used: ((((machine learning) OR (artificial intelligence) OR (deep learning) OR (Supervised learning))) AND ((stroke) OR (ICH) OR (intracerebral haemorrhage)). This returned articles dated between January 1st 1982 and December 1st 2024. Searches were limited to studies involving humans. No region or language limitations were applied.

2.2. Study selection

This review focused on studies evaluating the efficacy of ML in predicting the risk of future stroke from routine hospital data. This included demographics, past medical history, ECG, biochemistry, carotid doppler and physiological parameters.

Inclusion and exclusion criteria are summarised in Appendix 1, Table 6. Studies were confined to those that included human participants $\geq$18 years and used at least one ML to predict the future risk of haemorrhagic or ischaemic stroke. Since we focused on identifying articles to stratify stroke risk, case-control, cohort and randomised-clinical-control (RCT) studies were included, but cross-sectional studies were excluded due to their lack of temporal data.

Search results were uploaded to Rayyan software [19]. Duplicate studies were removed. Relevant studies were then identified by screening abstracts and titles against predefined inclusion and exclusion criteria. The full texts of the identified studies were then again screened against inclusion/exclusion criteria (Appendix 1). A selection of studies were then independently reviewed by two researchers. Relevant data and findings were manually extracted and presented in Tables 2, 3, 4 and 5.

3. Results

The initial search returned 8,677 references. Following removal of

Table 1
Descriptions of common machine learning models used in predicting stroke risk.

Models frequently used in predicting the risk of stroke from routine hospital data	
Logistic regression	Although familiar to many people, LR may be considered a type of machine learning, especially when used in combination with regularisation methods to improve prediction accuracy on unseen data. LR is a statistical model that models the probability of an event taking place by assuming that the log-odds for the event are a linear combination of one or more independent variables. LR is considered a simple and versatile method that is capable of illustrating clear relationships between variables. Because of its simplicity it is often used as a baseline tool in gauging variable importance for more complex models. However, LR may not be suitable for all datasets. Several assumptions are made, including a linear relationship between dependent and independent variables and no multicollinearity between variables. LR is also vulnerable to outliers and overfitting, making it less suitable for modelling complex datasets.[16]
Support machine vector learning	SVM is a technique most suited to classification of outcomes. SVM begins by graphically mapping data. These data are then labelled and segregated based on a given category. Data are separated via a plane, known as the hyperplane. Support vectors are data points that lie closest to the hyperplane. SVM identifies the optimal position of the hyperplane by maximising the margin (the distance between the hyperplane and the support vector). The wider the margin between the two categories, the better the fit. SVM is suitable for both linear and nonlinear cases. In non-linear cases, SVM classifier performs the transformation of input space data to a higher dimensionality using kernels for efficient classification of data. This added sophistication makes SVM better suited to modelling complex datasets where the number of variables outnumber the number of datapoints. The advantage of kernels allows mapping of data in higher dimensional spaces, enabling modelling of non-linear data. Furthermore, identification of a hyperplane with maximal margins between categories offers development of more generalisable models that are less vulnerable to overfitting. Because of their complexity SVM models can often be computationally expensive and time consuming making their application to large datasets limited. SVM has also been criticised for a lack of explainability and interpretability capability compared to simpler methods such as regression. [16]
Decision Tree	*DT* are popular models used in prediction problems. The purpose of DT is to predict the category or value of an outcome by learning numerous simple decision rules. DT begins with a "root node" that represents the entire dataset, which is subsequently divided into two or more homogenous sets known as decision nodes. These decision nodes are then further divided into either further decision nodes or terminal nodes. Terminal nodes represent the likely final outcome. To calculate this, DT use a variety of algorithms (such as regression or measures of entropy) to split the nodes for all available variables. A greedy search approach is employed to identify the optimal split points within a tree. DT are particularly useful in handling combinations of categorical and numerical data without the need for data transformation or making assumptions on data characteristics. They are also able to handle non-linear relationships between variables and perform variable selection, placing important variables at the top of the tree, improving generalisation. DT is limited when modelling large datasets, as models can be vulnerable to overfitting and instability. Because of their hierarchical approach, small changes in data can significantly alter tree structure. Furthermore, DT is often biased towards the class with the greatest number of datapoints, forming deeper branches in this class. [16]
Random Forest	*RF* attempts to address the above limitations of DT by adopting an ensemble approach. In an ensemble approach multiple DT models are generated, and their predictions are aggregated to identify the most popular result. The most commonly used method is bagging (bootstrap aggregation). In this method, a random sample of data is selected and undergoes replacement, meaning it can be selected more than once. Subsets datasets are then generated to build multiple DT. Where DT analysis must account for all possible variables, RF forces DT to randomly select a limited number of variables. Consequently, a large number of uncorrelated DT are generated, avoiding the issue of overfitting and increasing generalisation. However, as with many complex models, RF can be computationally and resource demanding. Due to the vast number of DT, Interpretability of the model is also limited making inference of results challenging. Gradient boosted algorithms, such as XGBoost aim to address these limitations. Gradient boosted algorithms minimise the number of trees needed by selectively adding trees which improve prediction (rather than random independent learners in RFs). Explainability is then added by using a SHAP model build on top of the gradient boosted trees.[16]
Neural networks	Neural networks are comprised of an input layer, one or more hidden layers and an output layer. Each neuron or node represents a mathematical function that transforms the value of the input data. If this value exceeds a certain threshold, the neuron "fires" and the information is passed onto the next layer, otherwise no data is passed on. A neural network is referred to as "deep" if more than three layers in total exist (i.e., more than one hidden layer). The neurons in a neural network are assigned "weights" that influence the probability of a neuron firing. These weights can be adjusted during training to optimise the model's ability to categorise input data. This is done by a process called backpropagation, in which the final output of the model is compared against the desired output. The error calculated is then used to modify the weight of each node. As data flows from one layer to the next, the output from one layer becomes in the input for another, until the final output layer is reached. This is known as a "feedforward" neural network. Other types of neural networks exist for specific applications. These include Multi-layer perceptron (MLP), convolutional neural network (CNN) and recurrent neural networks, but are beyond the scope of this article. [53] By adopting strategies such as back propagation, NN are able to automatically structure variables in hierarchical importance and handle noisy data, making them generalisable to new unseen data. They are also considered highly adaptable, adjusting model architecture to a wide range of data types. However, NN is limited by its large data and computational requirements in training models, consequently requiring expert input. Due to its complexity, NN can be vulnerable to overfitting and often performs poorly on imbalanced datasets. As a result further input is often required to tweak models with additional techniques such as normalisation, regularisation and re-weighting. NN is also criticised for its black box nature, making interpretability of models challenging. [16]

duplicates and review of titles and abstracts, 49 original research articles evaluating the efficacy of artificial intelligence in predicting first time stroke from routine hospital data were identified. A PRISMA diagram outlining the study selection process can be seen in Fig. 1. Publication dates ranged from May-2015 to December-2024. Most studies compared multiple AI models in their analysis. In order of most to least frequent, the most commonly used models were random-forests (RF), Logistic regression (LG), support vector machine (SVM), Decision-tree (DT), gradient-boosted-decision-tree (GBDT) and deep-neural-networks (DNN).

3.1. Population samples

Details on population statistics can be found in Tables 2 and 3. Sample sizes ranged from 110 cases to 6,457,412 cases (median size

Table 2
Summary of papers attempting to predict the future risk of stroke from routine hospital data in the general population. (See below-mentioned references for further information.)

Article	Variables	Cases: controls	Cohort	Follow up time	Exclusion criteria	ML models used (highest performing model in bold)	Validation method	Outcome	Sensitivity (highest performing model)	Specificity (highest performing model)	Accuracy (highest performing model)	AUC (highest performing model)	Main findings	Explainability	Interpretability	Calibration
A. Orfanoudakil et al [23] May-20	Demographics, PMH, medications, blood test data, ECG	167:9503	Offspring and spouses of the Framingham study cohort. Age 40+ (data on age and sex was not reported)	10 years	Previous stroke/TIA	**Optimal Classification Trees,** LG, CART, RF, Xgboost	10-fold cross validation. Dataset split into 75% for training and 25% for testing. Model was then externally validated on an independent cohort	Ischaemic stroke	-0.899	0.402	0.745	0.740 (0.715–0.771)	ML models trained on EHR were able to accurately predict stroke and outperformed traditional stroke prediction tools, such as Framingham score and QRISK. ML also identified 23 novel stroke risk factors including t wave abnormality and haematocrit.	No - no explainability analysis performed.	Yes- variables used in the model were listed. Variables such as smoking, cardiovascular disease and diabetes were highlighted as important risk factors for stroke. 23 novel risk factors were also identified, including T-wave abnormalities and haematocrit.	Yes Based on Hosmer-Lemeshow testing, optimal classification trees showed good calibration against real work probabilities.
K. Arslan et al[66] Jul-16	Demographics, stroke history, blood test data	80:112	Turgut Ozal Medical Centre, Turkey Sex = 52.6% M Average age = 53.97 ± 21.38 years.	?	?	**SVM**, SGB, PLR	10-fold cross-validation	Ischaemic stroke	0.975 (0.912–0.997)	0.982 (0.936–0.998	0.979 (0.974–0.994)	0.978 (0.957–0.999)	ML and particularly SVM is highly effective are predicting ischaemic stroke risk from routine medical data.	Yes - variables ranked in order of importance and importance values calculated.	Yes- variables used in the model were listed.	None - No testing for good calibration against real world probabilities was reported.
A. Jamthikar et al[48] Oct-19	B-mode Ultrasound carotids, PMH, demographics, blood test data	Total 202	Japan Sex = 77.2% M Average age = 68.97±10.96 years. Age range 29-88	18 months	?	**AtheroRisk-Integrated ML system** vs AtheroRisk conventional ML system (both RF)	10-fold cross-validation	Ischaemic stroke	0.550	0.992	0.952	0.80 (0.77-0.84)	ML was able to accurately predict stroke risk. Furthermore, integrating carotid ultrasound images into ML models improves overall model performance at minimal extra costs.	No - no explainability analysis performed.	Yes- variables used in the model were listed.	None - No testing for good calibration against real world probabilities was reported.
C. Colak et al [64] May-15	Demographics, PMH, carotid ultrasound, blood test data	130: 167	Turgut Ozal Medicine Center, Turkey Sex = 53.5% M Average age for men = 59.4 ± 16.3 years Average age for women = 63.3 ± 14.4 years	?	?	**ANN** and SVM	Dataset split into 70% for training and 30% for testing.	Stroke	-	-	0.859	0.928	ML was effective at accurately predicting stroke risk. Compared to SVM, ANN showed greater performance in predicting stroke.	Yes - variables ranked in order of importance and importance values calculated.	Yes- variables used in the model were listed.	None - No testing for good calibration against real world probabilities was reported.
C.Y Hung et al [20] Jul-18	demographics, PMH, DH	4795: 548103	Taiwan National Health Insurance Research Database Age range = 25 - 85 years (data on sex was not reported)	5 years	Previous stroke/TIA	**ADA,** SDA, TGL, MGL	5-fold cross validation. Dataset was split into 80% for training and 20% testing	Ischaemic stroke	-	-	-	AUC 0.805	Using EHR of any age group to augment ML models improves overall model performance. ADA outperformed other ML models in predicting young stroke	No - important variables not identified. no discussion on clinical explainability.	No - variables not identified	None - No testing for good calibration against real world probabilities was reported.
C.Y Hung et al[35] Mar-19	demographics, PMH, DH	2060: 670154	Taiwan National Health Insurance Research Database Sex = 48.5% male Average non-stroke age = 35.5±20.2 years. age range 25-84	3 years	Previous stroke	**DNN**	Dataset was split into 70% for training, 20% for testing and 10% for tuning.	Ischaemic stroke	0.925	0.798	-	0.920 (0.908–0.932)	DNN is able to accurately assess future stroke risk from EHR databases	No - important variables not identified. no discussion on clinical explainability.	No - variables not identified	Yes - The model did not calibrate well with real life probabilities (Hosmer-Lemeshow test = p<0.001), Platt scaling =0.039).
C.S Nwosu et	Demographics,	548: 28524	Kaggle	?	?	**Neural network,**	Dataset was	Stroke	-	-	0.752	-	ML, and in	No - no	Yes- variables used	None - No

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Table 2 (*continued*)

al[24] Jul-19	PMH, BM		(unspecified) (data on age and sex was not reported)			**RF**, DT	split into 70% for training and 30% for testing.						particular neural networks were effective at accurately predicting stroke risk from EHR.	explainability analysis performed.	in the model were listed.	testing for good calibration against real world probabilities was reported.
D. Teoh[88] Dec-18	demographics, PMH, blood test data	2725: 5450	Tsuyama Jifukai Tsuyama Chuo Hospital, Japan Age range = 45-95 years (data on sex was not reported)	?	Previous stroke/TIA	**DNN**	Dataset split into 70% for training, 15% for validation and 15% for testing.	Ischaemic stroke	-	-	-	0.669	DNN is able to effectively assess future stroke risk from EHR databases	Yes - variables ranked in order of importance and importance values calculated.	Yes- variables used in the model were listed. 3 most important variables identified and discussed (blood count, platelet count and diabetes).	None - No testing for good calibration against real world probabilities was reported.
E. Dritsas et al [25] Jul-22	demographics, PMH, blood test data	180: 3074	Kaggle (unspecified) Sex = 38.7% male Age = 18+ (data on age was not reported)	?	?	**stacking**, RF, LG, K-nearest neighbour, stochastic gradient descent, DT, multilayer perception, majority voting, naïve bayes	10-cross validation	Stroke	0.974	-	0.98	0.989	ML was able to accurately predict stroke risk, particularly when individual models were combined via stacking.	Yes - variables ranked in order of importance but no importance values given.	Yes- variables used in the model were listed	None - No testing for good calibration against real world probabilities was reported.
E. M Alanazi et al[65] Dec-21	demographics, PMH, blood test data, dietary data, physical exam	608: 3578	National Health and Nutrition Examination Survey Sex = 36.2% M, 31.3% F, not reported 198 (32.6%) Median age = 51	4 years	?	RF, DT, bayes net, naïve bayes	10-fold cross validation	Stroke	0.97	0.96	0.96	0.97	RF was highly accurate and easy to use in predicting the future risk of stroke	Yes - variables ranked in order of importance and importance values calculated.	Yes- variables used in the model were listed High lymphocyte and neutrophil and low folate counts were identified as novel predictors for stroke.	None - No testing for good calibration against real world probabilities was reported.
G. Lip et al [89] May-21	Demographics, PMH	Total 3,435,224	National private and Medicare insurance records, USA Age range = 45-95 years (data on sex was not reported)	2 years	?	LG, neural network, DT, gradient boosting)	Dataset was split into 2/3^{ds} for training and 1/3rd for validation.	Stroke including ischemic stroke, transient ischemic attack, and other thromboembolic events	-	-	-	c index 0.866 (0.856–0.876)	ML was able to accurately predict stroke in multimorbid populations and outperformed traditional tools such CHADSVASC and the clinical multimorbid index	Yes - variables ranked in order of importance and importance values calculated.	Yes- variables used in the model were listed	Yes - Based on decision curves, the model showed good calibration against real world probabilities.
G. Lip et al [57] Jun 2023	Demographics, PMH	6573: 147,978	Medicare health plan, USA. Age range = 18 - 94. Sex = 38% M	?	Previous heart disease, AF, stroke or TIA	**DNN**	Dataset was split into 2/3^{ds} for training and 1/3rd for validation.	Ischaemic stroke	-	-	-	0.6355	ML produced a model of good fit and accurate discriminatory performance for overcall cardiovascular risk and risk of ischaemic stroke.	Yes - Variables ranked in order of performance and odds ratios calculated.	Yes- variables used in the model were listed. Traditional factor such as diabetes, hypertension and previous stroke were identified as important risk factors, in addition to several putative risk factors including sleep apnoea and markers of frailty, such as osteoarthritis.	Yes - Based on platt curves, the model showed good calibration against real world probabilities.
J. Chen et al[53] Jan-22	Demographics, PMH, blood test data	?	Unknown location Average age = 54.1 age range = 27-89 years	?	?	**Novel Hybrid Deep Transfer Learning**, SVM, DT, RF and DNN	Dataset was split into 80% for training, 10% for validation and for 10% testing.	Stroke	0.715+/- 0.039	-	0.757 +/- 0.032	0.834 +/- 0.038	The authors propose a novel deep learning model capable of securely handling, pre-processing and balancing patient health data to accurately predict stroke risk.	No - no explainability analysis performed.	Yes- variables used in the model were listed.	None - No testing for good calibration against real world probabilities was reported.
J. Liu et al[36]	demographics, PMH, blood test data, dietary	135: 4246	National Health and Nutrition Examination	3 years	>30% unfilled questionnaire	**H2O Driverless AI**, Isolation forest, Cost-Sensitive	Dataset was split into 70% for training	Stroke	-	-	-	0.804 (0.770–0.838)	ML and in particular, H2O driverless AI was	Yes - variables ranked in importance	Yes- variables used in the model were listed.	None - No testing for good calibration
Sep-22	data, physical exam		Survey 2015–16, UK Sex = 48.8% male Average age = 63.89 ± 14.39 years			Neural Network ,weighted LG.	and30% for testing. Model was then validated on an external dataset						able to accurately predict stroke risk and identify several novel risk factors for stroke. A clinical dataset of PMH was deemed the most useful in predicting stroke, followed by blood test data and finally dietary data.	but no importance values calculated.	Models trained on different datasets were compared. Blood test and dietary data ranked as the least important, whilst PMH/ clinical data ranked as the most important. Several novel risk factors for stroke, such as asthma and snoring were identified.	against real world probabilities was reported.
HW. Chang et al[43] Nov-2023	Demographics, PMH, lifestyle, blood test data, ECG, physiological data	42:2166	Rugao Longitudinal Ageing Study, China. Mean age = 78.0 ± 4.4 years Age range 62-97 years Sex = 47.2% M	8 years	?	**SVM,** LR, RF, MLP, K-Nearest Neighbor, GBDT	10-fold cross validation. Dataset split into 75% for training and 25% for testing.	Ischaemic stroke	-	-	-	0.79 (0.77–0.81)	The use of ML improves stroke risk prediction compared to traditional models, such as LR. SVM appeared to be particularly useful in predicting stroke risk.	Yes - variables ranked in importance but no importance values calculated.	Yes- variables used in the model were listed. Several putative risk factors were identified, including heart rate variability, waist circumference and hypothyroidism.	Yes – calibration was performed but method or results were reported or discussed.
J. You et al[42] Apr-23	Demographics, PMH, DH, blood test data, FH, physical exam	5611: 442153	UK biobank, United Kingdom. median age = 57 years. Sex = 44.2% M	10 years	previous stroke or myocardial infarction.	**Light gradient boosting machine**, eXtreme Gradient Boosting Machine, RF, LG, K-nearest neighbours, SVM , artificial neural network.	20-fold cross validation. Dataset split into 95% for training and 5% for testing.	Ischaemic stroke	0.710 (0.649 - 0.771)	0.643 (0.637 - 0.650)	0.644 (0.582 - 0.706)	0.730 (0.710 - 0.750)	ML outperformed traditional scores such as QRISK in predicting cardiovascular risk including stroke.	Yes - variables ranked in importance but no importance values provided.	Yes- variables used in the model were listed. The model was able to accurately predict stroke from only 10 variables.	Yes – Based on calibration curves and Brier scores, models showed good calibration against real world probabilities ("Brier score for stroke = 0.057±0.006)
M. Khodadadi et al[63] Dec-19	demographics, PMH, blood test data, physiological data	Total 110	Amiralmomenin Hospital, Iran Age range = 28-95 years (data on sex was not reported)	5 years	?	**fuzzy cognitive mapping**, SVM, K-nearest neighbour	10-fold cross validation	Stroke	-	-	0.936 +/- 0.450	-	Fuzzy cognitive mapping is a unique ML approach towards stroke prediction and was superior to SVM and KNN in predicting the future risk of stroke. The FCM results were highly similar to the viewpoints of neurologists in identifying individuals at risk of stroke.	No - no explainability analysis performed.	Yes- variables used in the model were listed.	None - No testing for good calibration against real world probabilities was reported.

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Table 2 (*continued*)

M. Chun et al[37] Jul-21	Demographics, PMH, blood test data	43234: 460608	China Kadoorie Biobank 2004 - 2008, China Sex (stroke) = 45.3% M Sex (non-stroke) = 40.7% M Average age = 51.9 ± 10.6 years. Age range = 35-74	9 years	Previous stroke/TIA	**Cox regression, gradient boosted trees,** Framingham stroke risk profile, LG, SVM, random survival forests, , multilayer perceptron	k-fold cross validation. Dataset was split into 85% for training, 12.75% for validation and 2.25% for testing.	Ischaemic stroke	Men: 0.76 (0.72- 0. 80). Women 0.67 (0.64-0.71)	Men: 0.76 (0.75-0.78). Women 0.81 (0.80-0.82)	Men: 0.76 (0.75-0.77). Women 0.80 (0.79-0.81)	-	Ensemble, GBT and cox regression were the best performing models in predicting stroke. Ensemble methods provide an incremental benefit in identifying individuals high risk of stroke.	Yes - variables ranked in order of importance but no importance values calculated.	Yes- variables used in the model were listed. Many important variables included physical parameters such as blood pressure, weight and height.	Yes- Based on Hosmer-Lemeshow and Nam-D'Agostina testing, GBT demonstrated good calibration against real world probabilities.
M. Chun et al[38] Sep-21	Demographics, PMH, blood test data	644: 13109	China Kadoorie Biobank 2004 - 2008, China Sex (stroke) = 41.4% M Sex (non-stroke) = 37.4% M Age range = 35-74 years	3 years	Previous stroke/TIA	**sex-specific gradient boosted tree** vs cox models	3-fold cross validation. Dataset split into 85% for training and 15% for testing.	Ischaemic stroke	-	-	-	Men: 0.811 (0.753–0.867), Women: 0.743 (0.681–0.798)	ML was marginally better than Cox regression in predicting stroke. There was no benefit in building models of subsequent visits/ using temporal data, hence the authors advise that data should be collected at a single visit.	Yes - variables ranked in order of importance and importance values calculated.	Yes- variables used in the model were listed.	Yes- Based on Hosmer-Lemeshow, isotonic regression and Nam-D'Agostina testing, GBT demonstrated good calibration against real world probabilities.
S. Lolak et al [40]. Jul 2023	Demographics, PMH, DH, blood test data	9659: 265,588	Patients from Ramathibodi hospital, Thailand. Mean age = 65 years. Sex = 38% M)	10 years	previous stroke	**Extreme gradient boosting**, Naive Bayes, explainable boosting machine, Bayesian network, LG	Dataset split into 80% for training and 20% for testing.	Stroke	-	-	-	0.89 (0.88-0.90)	ML successfully predicted risk of stroke and identified important risk factors. Extreme gradient boosting was superior to other ML methods in predicting stroke.	Yes - variables ranked in order of importance and importance values calculated.	Yes- variables used in the model were listed. Important factors included traditional modifiable risk factors such as hypertension and high cholesterol.	None - No testing for good calibration against real world probabilities was reported.
S. Kurvits et al [61] Mar 2023	Blood blood test data and urine analysis.	950: 3800	Estonian biobank, Estonia. Mean age = 71 years. Sex = 41% M	16 years	?	**Ensemble of RF and DNN . LG**, K-nearest-number, RF, DNN .	10-fold cross validation. Dataset was split into 90% fortraining and. 10% for testing.	Ischaemic stroke	-	-	-	0.95	ML was successfully able to accurately predict the risk of ischaemic stroke from blood and urine blood test data. Several novel risk factors were also identified.	Yes - variables ranked in order of importance and importance values calculated.	Yes- variables used in the model were listed. Important putative risk factors included urine pH, haematocrit and RBC width,	None - No testing for good calibration against real world probabilities was reported.
T. Liu et al[54] Nov-19	Demographics, PMH, BM	783: 42617	Kaggle (unspecified) Age range = 0.08 - 82 years (data on sex was not reported)	?	?	**Bespoke DNN,** DNN, Bagging, RF, Adaboost, and XGBoost(XGB	10-fold cross validation	Stroke	0.674 +/- 0.05	0.326 +/- 0.05	0.716 +/- 0.12		The authors propose a unique pipeline where DNN is used to fill missing values within an imbalanced dataset and subsequently accurately predict stroke risk. This method was able to effectively reduce false negative rate, whilst maintaining high overall accuracy.	Yes - variables ranked in order of importance but no importance values calculated.	Yes- variables used in the model were listed. Important factors included traditional modifiable risk factors such as hypertension and BMI.	None - No testing for good calibration against real world probabilities was reported.
X. Li et al[26]	demographics,	206164:	China national	5 years	Previous stroke/TIA	**Boosting with C4.5**	10-fold cross	High stroke risk	0.999	-	-	B-DT: 0.999	Both boosting with	No - no	Yes- variables used	None - No

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Table 2 (*continued*)

Dec-19	PMH, FH blood test data	541350	stroke screening scheme, China Age = 40+ range (data on age and sex was not reported)			**DT,** voting, bagging with c4.5 DT, RF, neural network, DT, Bayesian network, naïve Bayesian, LG	validation. Dataset was split into 70% for training and 30% for testing.					[0.9991 – 0.9997) RF: 0.9941 (0.9939 - 0.9943)	DT and RF were highly effective at predicting future stroke risk. The authors felt that models would likely improve current screening methods by avoiding the impact of unknown values and avoid unnecessary rescreening and interventions.	explainability analysis performed.	in the model were listed.	testing for good calibration against real world probabilities was reported.
X. Li et al[62] Oct-22	demographics, PMH, blood test data	15833: 16,533	First Hospital of Jilin University, China Sex (stroke) = 69% M Sex (non-stroke) = 55% M Average age (stroke) = 58.0 years. Age range = 49.0–66.0 Average age (non-stroke) = 63.0 years. Age range = 55.0–71.0	?	Vital organ disease, neurological disease, infection, cancer, haematological disease. Additionally for controls - abnormalities on CT/MRI, digestive disease, metabolic disease, rheumatic disease, HTN, pregnancy, recent surgery, recent exercise.	**Extreme gradient Boosting (XgBoost**), LG (LR), SVM (SVM), and RF (RF)	5-fold cross-validation. Dataset was split into 70% for training and 30% for validation.	Ischaemic stroke	0.740	0.904	0.826	0.912 (0.905–0.919)	Extreme gradient boosting accurately predicted stroke risk from blood test data and outperformed LG.	Yes - variables ranked in order of importance but no importance values calculated.	Yes- variables used in the model were listed. Fibrinogen was identified as an important novel risk factor for stroke. The authors hypothesis that high levels of fibrinogen may reflect endothelial dysfunction seen in hypercoagulable and inflammatory states.	Yes- calibration Platt curves showed excellent calibration of Xgboost with real world probabilities..
Y. Wu et al[27] Mar-20	demographics, PMH, blood test data	56: 1075	Chinese Longitudinal Healthy Longevity Study 2012-2014, China Age = 60+ years (data on age and sex was not reported)	2 years	?	**regularised LG** (RLR), SVM (SVM), and RF (RF)	10-fold cross-validation. Dataset was split into 90% for training and 10% for testing	Stroke	0.75 (0.72–0.78)	0.69 (0.67–0.71)	0.70 (0.68–0.72)	0.72 (0.71–0.73)	ML techniques were able to accurately predict stroke risk, but only when balancing techniques such as SMOTE, RUS and ROS were employed.	Yes - variables ranked in order of importance but no importance values calculated.	Yes- variables used in the model were listed. Sex, hypertension and uric acid were considered important predictors of stroke.	None - No testing for good calibration against real world probabilities was reported.
Y. Liu et al[28] Sep-20	Demographics, PMH, transcranial doppler, ECG, CO2 and respiration during various movement task, 24-hour ECG and BP, EMG, accelerometer recordings, gait analysis.	60:60	cerebral vasoregulation in elderly with stroke data set, Israel (data on age and sex was not reported)	?	?	CNN **model based** (VGG19, DenseNet201, ResNet50, **VGG16**)	k-fold cross validation. Dataset was split into 60% for training, 20% for validation and 20% for testing	Occurrence probability of ischemic stroke	0.985	-	0.985	0.999	CNN methods were able to accurately predict the risk of stroke by combining multi-modal datatypes.	No - no explainability analysis performed.	Yes- variables used in the model were listed. The authors devised a bespoke pipeline to extract features from streaming data (e.g. heart rate and blood pressure) and combine this with PMH and demographics.	None - No testing for good calibration against real world probabilities was reported.
Y. Q iu et al [41] Mar 2023	Demographics, PMH, blood test data	31880: 14360	National stroke centre database, Jiangxi, China. Mean age = 62 years. sex =46% M	8 months	<20 years old. Previous stroke	**Extreme gradient boosting,** LG, DT, RF, gradient boosting DT.	10-fold cross validation. Dataset was split into 75% for training and 25% for validation.	Stroke	0.776	0.916	0.873	0.924	ML is able to provide accurate stroke risk prediction. RF and XGBoost were slightly superior to LG and less prone to overfitting.	Yes - variables ranked in order of importance but no importance values calculated.	Yes- variables used in the model were listed. Physical inactivity, hypertension and diets high in salt and meat were considered important risk factors. Homocysteine was also identified as a novel risk factor.	Yes- calibration curves for all models showed almost perfect calibration to real world probabilities
Y. Zhang et al [90] Apr-19	demographics, PMH, blood test data	398:394	Unspecified community hospital, China Sex = 41.9% M Age range = 45-90 years	5 years	?	**SVM,** Bayes, CBA, BPNN, CART, C4.5, ELM	10-fold cross validation	Stroke	0.827	0.804	0.815		The authors devise a novel feature selection method, WRHFS, which can be combined with ML models (namely SVM) to accurately predict the future risk of stroke.	Yes - variables ranked in order of importance but no importance values calculated.	Yes- variables used in the model were listed. Several novel risk factors, including α Hydroxybutyric dehydrogenase and Gamma glutamyl transpeptidase were considered important predictors of stroke.	None - No testing for good calibration against real world probabilities was reported.
Y. Yang et al [39] Nov-21	demographics, PMH, blood test data, physiological data	8827:42088	Shenzhen Health Information Big Data Platform, China Sex = 59.5% M Average age (stroke) = 59.7 ± 11.9 years Average age (non-stroke) = 60.2 ± 11.8 years Overall age range = 35-64 years	3 years	SAH, ICH, previous stroke, absence of follow-up records. Additionally for controls - history of heart disease, renal disease or cancer.	**Extreme gradient boosting,** LG, SVM, RF	5-fold cross validation. Dataset was split into 70% for training and 30% for testing.	Stroke	0.851	0.845	0.850	0.922	Extreme gradient boosting provided an accurate 3-year stroke prediction model in hypertensive patients. This outperformed traditional risk scales, such as the Framingham criteria and QRISK	Yes - variables ranked in order of importance but no importance values calculated.	Yes- variables used in the model were listed.	None - No testing for good calibration against real world probabilities was reported.
P. Chakraborty Oct 2024[33]	Demographics, PMH	Total 5110	Undefined Kaggle dataset (data on age and sex was not reported)	?	?	**Stacked ensemble model (KNN, DT, RF)** KNN, DNN, RF, SVM, ADA, MNB	5-fold cross validation. Dataset was split into 80% for training and 20% for validation.	Stroke	0.996	-	0.986	-	ML was able to accurately predict stroke risk by leveraging a principal component analysis and stacking ensemble approach.	No - no explainability analysis performed.	Yes- variables used in the model were listed.	None - No testing for good calibration against real world probabilities was reported.

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J. Zheng Aug 2023[91]	demographics, PMH,	113: 120	Chengdu and Chongqing, China. (data on age and sex was not reported)	?	?	**RF,** LR, NB, DT, KNN, SVM	Dataset was split into 75% for training and 25% for testing.	Stroke	0.762	-	0.755	-	ML was able to accurately identify individuals' high risk of stroke. RF and KNN performed better than other models in predicting stroke risk.	No, but Chi square and multivariate logistic regression performed for variables vs stroke	Yes- variables used in the model were listed. Traditional risk factors such as age, hypertension, AF and drinking were identified as important predictors of stroke.	None - No testing for good calibration against real world probabilities was reported.
M. Aboonq, S. Alqahtani Jul 2024[55]	demographics, PMH	17,214 :421,479	2021 Behavioral Risk Factor Surveillance System, USA. Sex = 54% M, largest age group = 65-69years, age range= 18 – 80+	?	?	**RF**, Xgboost, DNN, LR, KNN	5-fold cross validation	Stroke	0.775	-	0.725	0.725	ML was able to predict stroke risk from routine data and identify important risk factors. RF was superior to other models in predicting stroke risk.	Yes - variables ranked in order of importance but no importance values calculated.	Yes- variables used in the model were listed. Traditional risk factors, such as hypertension, ischaemic heart disease and age were considered important predictors of stroke.	None - No testing for good calibration against real world probabilities was reported.
U. Islam et al Sep 2024[32]	demographics, PMH	465: 4645	Undefined	?	?	**Ensemble model (KNN, LR, SVM, NB),** KNN, LR, SVM, NB	k-fold cross validation	Stroke	0.97	-	0.96	-	ML was able to accurately predict risk of future stroke, especially when models were combined using nested ensembling.	No - no explainability analysis performed.	Yes- variables used in the model were listed.	None - No testing for good calibration against real world probabilities was reported.
C. Yang et al Aug 2024[92]	demographics, PMH, blood test data.	150:150	Fushun Central Hospital and Yancheng Third People's Hospital, China. Sex = 71% M, Average age (stroke) = 66 years Average age (non-stroke) = 63 years	?	Exclusion criteria for stroke: infectious disease within 2–4 weeks; severe immune system diseases; malignant tumors; serious vital organ diseases; use of steroids or immunosuppressants; cerebral hemorrhage; use of anti-inflammatory drugs; hematological diseases; and inability to undergo MRI examination. Exclusion criteria for non-stroke patients: History of stroke; severe immune system diseases; severe infections; and malignant tumors.	**LR,** lightGBM, CNB, SVM, MLP	10-fold cross validation. Dataset split into 90% for training and 10% for testing	Acute cerebral infarction	0.899 (0.860–0.937)	0.828 (0.780–0.877)	0.833 (0.825–0.841)	0.913 (0.880–0.947)	ML, and in particular logistic regression was effective at accurately predicting stroke risk from routine blood test data.	Yes - variables ranked in order of importance but no importance values calculated.	Yes- variables used in the model were listed. Glucose, sex and smoking history were amongst the most important risk factors for stroke. Several novel risk factors including homocysteine, folic acid and CRP. The authors hypothesise this may be an indicator of underlying endothelial dysfunction, contributing towards atherosclerosis and stroke risk.	None - No testing for good calibration against real world probabilities was reported.

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D. Badalotti et al Dec 2024[93]	Demographics, PMH	?	IRCCS Humanitas Research Hospital, Italy. Sex = 51% M Median age = 75	7 years	Premorbid modified Rankin Scale (mRS) ≥ 3	**Natural language processing model - Bidirectional Encoder Representations from Transformers**	10-fold cross validation.	Ischaemic stroke	0.662	-	0.740	0.795	Natural language processing was able to predict stroke risk from unstructured health texts.	No - no explainability analysis performed.	Yes- variables used in the model were listed.	None - No testing for good calibration against real world probabilities was reported.
C. Gao, H. Wang Jan 2024[31]	Demographics, PMH	?	Undefined Kaggle dataset (data on age and sex was not reported)	?	?	**DNN,** DT, SVM, RF, ANN	Dataset split into 70% for training, 20% for testing and 10% for validation.	Stroke	-	-	0.976	-	The authors novel DNN model was superior to other ML models in predicting stroke. Wasserstein generative adversarial networks with gradient penalty was an effective strategy in addressing class imbalance and superior to SMOTE for this task.	No - no explainability analysis performed.	Yes- variables used in the model were listed.	None - No testing for good calibration against real world probabilities was reported.
T. Vu et al Jul 2024[94]	demographics, PMH, blood test data.	436: 6953	Suita City, Japan Sex = 46% M Average age = 56 years. Age range = 30-84	10 years	?	**RF,** Xgboost, LR, SVM, lightGBM	5 fold cross validation. Dataset split into 80% for training and 20% for testing.	Stroke according to U.S. National Survey of stroke criteria.	0.70 ± 0.05	-	0.70 ± 0.05	0.71 ± 0.06	Both supervised and unsupervised ML models were effective at accurately predicting stroke risk. RF appeared superior to other ML models in predicting stroke risk.	Yes - variables ranked in order of importance but no importance values calculated.	Yes- variables used in the model were listed. Important risk factors included age, hypertension and eGFR. Several novel factors were also identified, such as elbow thickness, serum fructosamine and serum calcium.	None - No testing for good calibration against real world probabilities was reported.
T. Tazin et al Nov 2021[30]	Demographics, PMH	249:4861	Undefined Kaggle dataset (data on age and sex was not reported)	?	?	**RF,** voting classifier, DT, LR.	Dataset split into 80% for training and 20% for testing.	Stroke	0.95	-	0.96	-	ML, and in particular RF was able to accurately predict stroke risk.	No - no explainability analysis performed.	Yes- variables used in the model were listed.	None - No testing for good calibration against real world probabilities was reported.

PMH – past medical history; ECG – electrocardiogram; DH – drug history; BP – blood pressure; TIA – trans ischaemic attack; HTN – hypertension; ICH – intracerebral haemorrhage; SAH – subarachnoid haemorrhage; SVM – support vector machine; CART – classification and regression tree; SGB – stochastic gradient boosting; PLR – penalised LG; ANN artificial neural network; SDA – stacked denoising autoencoders; ADA – adaptive boosting; TGL – target group learning; MGL – Multiple group learning; DNN – deep neural network; RF; LR – LG; Xg boost – extreme gradient boosting; RLR – regularised LG; CBA – classification based on association; BPNN – back propagation neural network; ELM – extreme machine learning; GBT – gradient boosted tree; AI – artificial intelligence; ML – machine learning; WRHFS – weighting- and ranking-based hybrid feature selection; FCM – fuzzy cognitive mapping; EHR – electronic health records, MNB – multinomial naïve bayes.

6844.5).

For most studies, the average age in the training cohorts was between 43 and 77 years (mean age 62.5 years). However, one study assessing young stroke had a mean age of 35.5 [20]. Fourteen studies did not provide descriptive age statistics [21–33]. Age range varied between studies but was often around 25–90 years, with a negatively skewed distribution. 27 of 49 studies were based in Asian populations and 13 in European or North American populations.

Overall, the observation period prior to the onset of stroke varied from 1 year to 16 years (mean follow up 5.1 years). Eighteen studies did not provide information about their observation period.

Differences between demographics in studies including populations with AF versus populations with or without AF are summarised in Appendix 1, Table 7. Notable differences were that the mean age for studies including populations with AF only was 66.9 years (range 43.9 – 77.6 years) versus 62 years (range 51–78 years) for studies including populations with or without AF. The median sample size for studies including populations with AF only was 6300 (range 173 – 6,457,412) and 7389 (110 – 3,435,224) for studies including populations with or without AF.

The majority of studies excluded patients if they had a previous TIA/stroke, although 23 studies did not specify their exclusion criteria.

Most studies used the international-classification-of-diseases-tenth-revision (ICD-10) criteria to define stroke as their primary outcome

Table 3
Summary of papers attempting to predict the future risk of stroke from routine hospital data in patients with AF.

Article	Variables	Case: Controls	Cohort	Follow up period	Exclusion criteria	ML models used (highest performing model in bold)	Validation	Outcome	Sensitivity (highest performing model)	Specificity (highest performing model)	Accuracy (highest performing model)	AUC (highest performing model)	Main findings	Explainability	Interpretability	Calibration
H. Nishi et al[50] Dec-21	Demographics, PMH, blood test data, echocardiography, chest-xray	129: 1628	Fushimi AF Registry, Japan Sex = 58.3% M Average age = 73 years.	6 years	Valvular AF, anticoagulation therapy	**GBDT (CatBoost)**, Cox proportional hazards	5-fold cross validation.	Ischaemic stroke	0.74 (0.59–0.87)	0.54 (0.48–0.59)	0.55 (0.50–0.60)	0.72 (0.66–0.79	ML was superior to CHADS2 and CHA2DS2-VASc in predicting the risk of ischaemic stroke in non-valvular AF patients.	No - no explainability analysis performed.	Yes- variables used in the model were listed. Left ventricular wall thickness (LVWT) was identified as an important variable. Authors hypothesise that hypertension and ischaemic heart disease may drive increased LVWT, which contributes towards AF pathogenesis. Hence, LVWT may be a marker of hypertension, heart disease and/or AF, all of which contribute towards stroke risk.	None -No testing for good calibration against real world probabilities was reported.
Y. Chen et al[44] Apr-24	Demographics, PMH, DH, blood test data, echocardiography	83:2018	KERALA-AF registry, Kerala, India. Median range = 68 years Sex = 53.8% M	1 year	Valvular AF, those lost to follow-up	**SVM**, LightGBM, RF, LR, MLP,	5-fold cross validation Dataset split into 70% for training and30% for validation. Models were then externally validated on APHRS-AF registry	Ischaemic stroke	0.692	0.919	0.912	0.835 (0.831, 0.839)	ML was superior to CHA2DS2-VASc in predicting the risk of stroke However, on external validation ML performed modestly with an AUC of 0.67	Yes - variables ranked in order of importance and variable importance values were calculated, to gauge the relative importance of each variable.	Yes- variables used in the model were listed. Authors discuss how age, prior stroke/TIA and enlarged left atrial size are important predictors of stroke risk.	None -No testing for good calibration against real world probabilities was reported.
J. Lu et al[58] Nov-22	Demographics, PMH, DH, blood test data	9670	AF patients from Sir Charles Gairdner and Osborne Park teaching hospitals in Perth, Western Australia Sex = 43.1% M Average age = 76.9 (SD.12.1) years	1 year	Mitral valve stenosis, prosthetic heart valve	**Multilabel GBM,** classifier chain, SVM, and MLNN	Dataset split into 90% for training and for 10% testing.	ICD-10 diagnosis of ischaemic stroke Major bleeding	0.976	-	-	0.685 (0.676, 0.694)	Multilabel ML models were superior to clinical risk stratification scores (e.g. CHA2DS2-VASc and HAS-BLED) in predicting the risk of major bleeding, stroke and all-cause mortality in non-valvular AF patients.	Yes - variables ranked in order of importance and variable importance values were calculated, to gauge the relative importance of each variable.	Yes- variables used in the model were listed. Key variables, such as age and previous stroke/TIA were highlighted, however many variables were non-specific and did not reflect true disease processes (e.g. No. hospital admissions).	Yes – brier scores showed that multilabel GBM was well calibrated against real world probabilities and continued to outperform tradition risk scoring tools such as CHA2DS2-VASc.
K. Kostev et al[49] May-21	Demographics, PMH	2310: 32342	Disease Analyzer database (IQVIA), Germany. Sex = 45.6% M (stroke), 50% M (non-stroke) Average age = 77.6 years (SD 8.8) (stroke), 75.7 years (SD 10.3) non-stroke	7 years	previous stroke/TIA	**SubPopulation Optimization and Modelling Solution** - **SOMS**(Classification and Regression Trees)	10-fold cross validation	TIA/Stroke	-	-	-	-	For patients with AF treated with a NOAC, SOMS was able to accurately predict stroke and identify several new risk factors for the future risk of stroke.	Yes - variables ranked in order of importance and variable importance values were calculated, to gauge the relative importance of each variable.	Yes- variables used in the model were listed. Urinary tract infection identified as an important novel risk factor, highlighting how infection may acutely increase stroke risk.	None - No testing for good calibration against real world probabilities was reported.

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L Han et al[60] Oct-19	Demographics, PMH, DH, cardiac monitoring/ECG data	71:3114	identified Veterans Health Administration, USA Sex = 98.6% M (stroke) 98.5 % M (non-stroke) Average age = 69.0 ± 11.0 years (stroke) 67.6 ± 10.3 years (non-stroke)	30 days prior to stroke	Oral anticoagulation, previous stroke/TIA.	**CHA2DS2-VASc with RF and CNN ensemble,** CNN (CNN), RF (RF), and L1 regularised LG (LASSO).	Dataset split into 50% for training, 20% for validation, and 30% for testing.	stroke	-	-	-	0.696	ML and AF signature burden was able to accurately predict stroke risk and outperformed CHA2DS2-VASc in predicting 30 day stroke risk.	Limited – explainability analysis was lacking and no values for variable importance were calculated in order to compare to importance between variables.	Yes- variables used in the model were listed. The authors highlight that the daily burden of AF was an important predictor of stroke but did not theorise on why this might be.	Yes – models were tested for calibration against real world probabilities using Platt scaling, however the results of this was not reported.
R. Leonarduzzi et al[29] May-18	24-hour Holter monitors (ECG data)	22: 151	Fujita Health University Hospital, Japan (data on age and sex was not reported)	47 months	<1 year of AF, Complete AV block, sustained ventricular tachycardia, ventricular ectopy>5%, cardiac pacemakers, paroxysmal, heart valves, with more than 5% of the Holter record corrupted by artefacts or noise, taking rhythm-control drugs, or that had acute coronary syndrome, hemodynamic instability, surgery within last 6 months. Stroke/TIA	**Sparse SVM**	5-fold cross-validation. Dataset split into 70% for training and 30% for validation.	Ischaemic stroke	-	-	-	0.65	S-SVM trained on ECG data was superior to CHA2DS2-VASc in predicting the future risk of ischaemic stroke. Variation in heart rate was identified to be an important risk factors for ischaemic stroke, particularly in those not anticoagulated.	Limited – explainability analysis was lacking and variables were not clearly defined.	Yes - variables reported but were not clearly defined. Although variation in heart rate was identified as a potential novel risk factor for stroke, there was no discussion on how this or other variables may contribute towards stroke risk	None - No testing for good calibration against real world probabilities was reported.
S. Jung et al[21] Nov-21	Demographics, PMH, DH, blood test data, physical observations	62226: 692723	Korean National Health Insurance Service database, Korea Sex = 49.7% M (stroke) 59.2% M (non-stroke) Age = 40+ years	5 years	Previous stroke/TIA	**DNN**, LG, XG boost, RF	10-fold cross validation. Dataset split into 60% for training, 20% for validation and 20% for training.	ICD-10 diagnosis of ischaemic stroke	0.764+/-0.005	-	-	0.727 ± 0.003	DNN was superior to CHA2DS2-VASc score in predicting the future risk of stroke in AF patients. Electronic-health-records were a useful resource in building such models.	Yes - variables ranked in order of importance and importance values calculated in supplementary material	Yes- variables used in the model were listed. Authors primarily focused on feasibility on implementing EHR with ML models	None - No testing for good calibration against real world probabilities was reported.
X. Li et al[22] Feb-17	Demographics, PMH, DH, blood test data	193: 1671	Chinese Atrial Fibrillation Registry, China (data on age and sex was not reported)	2 years	?	**LG**, Cox, Naïve Bayes, CART, RF	10-fold cross validation. Dataset split into 60% for training and 40% for testing.	Ischaemic stroke	-	-	-	0.744	ML was superior CHA_2DS_2-VASc and the Framingham risk score in predicting ischaemic stroke (AUC: 0.66-0.69). ML was also able to identify new potential risk factors for stroke.	Limited – explainability analysis was performed but details on feature importance was not reported.	Yes- variables used in the model were listed. The authors highlight several novel risk factors for stroke, such as diabetes exposure, cardiovascular disease durations and certain ECG parameters, However, these were not defined in detail.	None - No testing for good calibration against real world probabilities was reported.
G. Lip et al[51] May-21	Demographics, PMH, AF exposure time	128047: 6329365	US Medicare claims, USA Sex = 56.6% M Average age = 43.9 (SD 15.8) years	53 months	Previous stroke/TIA	**LG**, neural network, DT, gradient boosting)	Dataset split into 2/3rds for training and1/3rd forvalidation. The model was then validated on an external dataset	ICD-10 diagnosis of ischaemic stroke	, -	-	-	(C index 0.891, 95% CI 0.882–0.9)	ML and multi-morbid index were superior to CHADS2 and CHA2DS2-VASc scores in predicting the future risk of stroke in AF. The model also performed well on external validation (AUC 0.891)	Yes - variables ranked in order of importance and importance values calculated.	Yes- variables used in the model were listed. Risk factors such as AF exposure time, time since previous stroke and age were identified as important risk factors.	Yes – Models were tested for good calibration using unspecified graphical methods and demonstrated "good internal and external calibration".
A. Papadopoulou et al Mar 2024[45]	Demographics, PMH, DH, biochemistry, genetic data	3150:3150	UK-biobank, UK Sex = 53% M, Mean age = 59 years (std. 8 years) Age range = 37 – 73 years	4 years	Previous stroke/TIA, previous diagnosis of cardiac arrythmias, previous cardiac surgery.	**Xgboost,** lightGBM, RF, SVM, DNN, LR	10-fold cross validation Data was split into unspecified training, test and validation sets.	ICD10 diagnosis of ischaemic stroke	0.63	-	0.63	0.631 (0.604–0.657)	ML appeared superior to CHA_2DS_2-VASc in predicting ischaemic stroke. The addition of routine blood test to stroke prediction models may also be of value.	Yes - variables ranked in order of importance and importance values calculated.	Yes- variables used in the model were listed. Creatinine was identified as an important predictor of stroke, Genetic data/ variants did not appear overly valuable in predicting stroke risk.	None - No testing for good calibration against real world probabilities was reported.
J. Lu et al Feb 2024[46]	Demographics, PMH, DH	215: 14480	GLORIA-AF registrar, data across 38 countries globally. Sex = 55.2% M Mean age = 70 years (Std 10 years).	1 year	Participants lacking adequate follow-up data	**Multi-label gradient boosted decision tree**	Dataset was split into 60% for training, 30% for testing, and 10% for validation.	Ischaemic stroke Major bleeding All-cause mortality	0.680 (0.558, 0.955)	0.618 (0.608, 0.629)	-	0.698 (0.589, 0.807)	Multilabel ML was superior to CHA_2DS_2-VASc and HAS-BLED in predicting ischaemic stroke, major bleeding and all-cause mortality.	Yes- variables ranked in order of importance, however only for all-cause mortality and not specifically for stroke.	Yes- variables used in the model were listed. Age, smoking, previous stroke/TIA and creatinine clearance were considered important predictors for stroke.	None - No testing for good calibration against real world probabilities was reported.

CXR – chest X-ray; PMH – past medical history; DH – drug history; VA –; AF – atrial fibrillation; USA – united states of America; SD – standard deviation; TIA – transient ischaemic attack; NOAC- new aged anticoagulation; SVM – support vector machine; GBM – gradient boosted machine; MLNN – Multi-layered neural network; CNN – convolutional neural network; LASSO – L! regularised logistic regression; RF – RF; DNN – deep neural network; XG boost – extreme gradient boosting; CART – Classification And Regression Trees for Machine Learning.

[34]. Twenty studies did not state the criteria they used to define stroke.

Thirty-eight studies assessed the risk of stroke in the general population (which included groups with and without atrial fibrillation), and eleven studies in patients specifically with atrial fibrillation.

Only 12 of 49 articles [35–46] reported the recommended minimum information on cohort demographics, exclusion/inclusion criteria and/or primary outcome as outlined by EQUATORs (enhancing quality and transparency of health research) "TRIPOD" checklist for prediction models [47].

3.2. Model performance

Tables 2 and 3 summarise the studies aiming to predict the future risk of stroke in the general population and in patients with AF respectively. All studies concluded that ML methods were able to accurately predict the risk of future stroke in the general population and in patients with AF, including thirteen articles that compared ML against clinical scoring tools and found ML to be more accurate/predictive. Several studies also highlighted the ML was useful in identifying new risk factors for stroke, which are highlighted below in Tables 4 and 5.

Table 4
Summary of variables used in machine learning models designed to predict the future risk of stroke in the general population.

A. Orfanoudaki et al 2020[23]	Variables (Total: 28)	Age	Gender	smoking	CVD	TIA	MI	BP Category	HTN medication	T-Wave abnormality	SBP
K. Arslan et al 2016[66]	Variables (Total: 18)	Age	Cr	Cl	Urea	Glc	Hb	K	Gender	INR	Htc
	Importance	100	73.827	72.159	68.061	60.984	49.19	44.174	28.902	28.568	25.268
C. Colak et al 2015[64]	Variables (Total: 8)	Carotid Doppler	HTN	cerebrovascular disease	CRP	DM	CH9D	Cholesterol	Smoking		
	Importance	0.23	0.13	0.06	0.2	0.09	0.03	0.17	0.09		
HW Chang et al 2023	Variables (Total: 103)	History if ischaemic stroke	MMSE	Homocysteine	SLDL-CLDL cholesterol	Folic acid	Triglycerides	Waist circumference	Total cholesterol	Age	Rv5 amplitude
	Importance	1	2	3	4	5	6	7	8	9	10
C.S Nwosu et al 2019[24]	Variables (Total: 10 of 12)	gender	age	HTN	Heart disease	marital status	occupation	Residence type	Glc	BMI	smoking
E. Dritsas et al 2022[25]	Variables (Total: 10)	Age	BMI	Glc	Ever married	Work type	Smoking	Residence type	Gender	Heart disease	HTN
	Importance	0.4702	0.404	0.1139	0.0929	0.0898	0.06618	0.0537	0.05	0.0499	0.0177
E. M Alanazi et al 2022[65]	Variables (Total: 20)	Age	Gender	RBC width (%)	lymphocytes (%)	RBC folate (ng/mL)	Segmented neutrophils	Hb (g/dL)	RBC count (million cells/microL)	Hct (%)	lymphocytes (1000 cells/ microL)
	Importance	0.26	0.13	0.18	0.15	0.13	0.12	0.11	0.11	0.09	0.08
G. Lip et al 2023 [57]	Variables (total: 28)	family history f stroke	alcohol dependance	cognitive impairment	dDM	depression	HTN	COPD	bleeding history	CKD	smoker
	Importance (OR)	1.64	1.51	1.4	1.26	1.25	1.25	1.24	1.23	1.2	1.17
G. Lip et al 2021 [88]											
J. Chen et al 2022 [53]	Variables (Total: 23)	Age	Neutrophils	lymphocytes	eosinophils	basophils	Total protein	Albumin	Globulin	Total bili	Direct bili
J. Liu et al 2022 [36]	Variables (Total: 278)	Age	Income	Smoking	Caucasian	Smoking	African American	HTN	DM	Asthma	Snoring
	Importance	1	2	3	4	5	6	7	8	9	10
J. You et al 2023 [42]	Variables (total: 645)	Age	Cystatin C	SBP	HTN	smoking pack years	fat free body mass	Sex	Microabluniria	Albumin	Long standing illness
	Importance	1	2	3	4	5	6	7	8	9	10
M. Khodadadi et al 2019[63]	Variables (Total: 12)	Age	BP	LDL	HDL	DM	heart disease	family history	smoking	BMI	exercise
M. Chun et al 2021[37]	Variables (Total: 25)	SBP	Age 65+	Insurance	Retired	Height	Harbin	Waist/hip ratio	Waist size	Age	Phys Activity
	Importance	0.3035	0.1399	0.1012	0.064	0.059	0.0454	0.0355	0.0339	0.0335	0.0315
M. Chun et al 2021[38]	Variables (Total: 133)	Age	SBP	SBP mean	met	weight	standing height	DBP mean	sitting height	BMI	Heart rate
	Importance	0.093	0.051	0.047	0.043	0.042	0.039	0.038	0.038	0.034	0.033
S. Lolak et al 2023 [40]	Variables (total: 19)	AF	Statin	Male	HTN	HTN medication	anti = platelet	Statin	Diabetes medication	age	SBP
	Importance	0.241	0.132	0.13	0.129	0.077	0.071	0.049	0.043	0.022	0.02
S. Kurvits et al 2023[61]	Variables (Total: 30)	RBC distribution width	Mean platelet volume	neutrophils	haematocrit	cholesterol	Hb	lymphocyte %	Creat	LDL	HDL
	Importance	1	2	3	4	5	6	7	8	9	10
T. Liu et al 2019 [36]	Variables (Total: 10)	Age	glucose	BMI	work	HTN	married	smoker	heart disease	gender	residence
	Importance	1	2	3	4	5	6	7	8	9	10

(*continued on next page*)

Table 4 (*continued*)

A. Orfanoudaki et al 2020[23]	Variables (Total: 28)	Age	Gender	smoking	CVD	TIA	MI	BP Category	HTN medication	T-Wave abnormality	SBP
X. Li et al 2019 [26]	Variables (Total: 17)	Sex	Age	Smoking	Exercise	Drinking	overweight	FH of stroke	FH of heart disease	FH of HTN	FH of DM
X. Li et al 2022 [62]	Variables (Total: 48)	Fibrinogen	Age	alb	NE	MCHC	Glc	Triglycerides	APTT		
	Importance	1	2	3	4	5	6	7	8		
Y. Qiu et al 2023 [41]	Variables (Total: 13)	Physical inactivity	HTN	meat diet	salt intake	high cholesterol	DM	sex	cardiac history	age	alcohol intake
	importance	1	2	3	4	5	6	7	8	9	10
Y. Wu et al 2020 [27]	Variables (Total: 15)	Sex	Alcohol	HTN	Glc	uric acid					
	Importance	1	2	3	4	5					
Y. Liu et al 2020 [28]	Variables (Total: 39)	HTN	age	Alcohol	Neuropathy	previous tobacco use	current tobacco use	Heart disease FH	previous stroke	AF	BMI
Y. Zhang et al 2019[89]	Variables (Total: 28)	Age	α-HBD	SCr	LDH	Height	Total − Bili	CK	Apo-B	CK-MB	Alb
	Importance	1	2	3	4	5	6	7	8	9	10
Y. Yang et al 2021 [39]	Variables (Total: 77)	Age	years after HTN	DBP mean	DBP delta mean	HR Mean	PPD Mean	SBP Mean	PDD delta mean	DBP	SBP delta mean
	Importance	1	2	3	4	5	6	7	8	9	10
M. Aboonq, S. Alqahtani 2024 [55]	Variables (Total: 10)	HTN	MI	angina	Diabetes	High cholesterol	gender	BMI	smoking		
	Importance	1	2	3	4	5	6	7	8	9	10
C. Yang et al 2024 [91]	Variables (Total: 36)	Glucose	Gender	Smoking	homocysteine	Folic acid	CRP	Total cholesterol	Age	Vitamin B12	BMI
	Importance	1	2	3	4	5	6	7	8	9	10
T. Vu et al 2024 [93]	Variables (Total: 53)	Age	SBP	HTN	eGFR	Metabolic syndrome	Glucose	Elbow	Fructosamine	Haemoglobin	Calcium
	Importance	1	2	3	4	5	6	7	8	9	10

BP – Bloods pressure; HTN – hypertension; SBP – systolic blood pressure; CRP – C-reactive protein; DM – diabetes mellitus; CHD – coronary heart disease; Glc – Glucose; RBC – red blood cell; Hb – haemoglobin; Hct – haematocrit. Bili – bilirubin; LDL – low density lipoprotein; HDL – high density lipoprotein; BMI – body mass index; FH – family history; alb – albumin; NE –; MCHC – Mean corpuscular haemoglobin concentration; APTT – Activated Partial Thromboplastin Clotting Time; α-HBD − Serum α-hydroxybutyric dehydrogenase; sCr – serum creatinine; LDH − Lactate dehydrogenase; CK – creatine kinase; Apo-B – apolipoprotein B; CK-MB − creatine phosphokinase-MB; MMSE – mini mental state exam.

Table 5
Summary of variables used in machine learning models designed to predict the future risk of stroke in patients with AF.

H. Nishi et al 2021 [50]	Variables (Total: 14)	Age	Height	Weight	Stroke	HTN	Antiplatelet	Antiarrhythmic drug	Creatine	Urea	Bleeding history	Mitral regurg
J. Lu et al 2022 [58]	Variables (Total: 43)	Previous stroke/ tia	Age	Cardiology admission	year of cohort entry	Hb	length of hospital stay	No. of comorbidities	Rate control drug	Bleeding history	eGFR	
	Importance	1	2	3	4	5	6	7	8	9	10	
K. Kostev et al 2021[49]	Variables (Total: 43)	Age	IHD	UTI	Dementia	Male	Dizziness/ giddiness	Dorsalgia	Shoulder lesion	DM		
	Importance	59.1	10.6	4.6	3.5	3.5	1.5	1.1	1.1	1.1		
Y. Chen et al[44]	Variables (Total	CKD	Age > 75	HTN	AF treatment	Elevated AST	Diuretics	female	diabetes	Prior CVA/TIA/ SE	Enlarged LA size	
	Importance	1	2	3	4	5	6	7	8	9	10	
S. Jung et al 2021 [21]	Variables (Total: 54)	Operation history	PE	Insurance type (employed)	Male	Cancer	CKD	Cardiac arrhythmias	Anaemia	Haemorrhoids	Hyperplasia of prostate	
	Importance (OR)	0.637 (0.592–0.685)	0.764 (0.663–0.88)	0.806 (0.78–0.833)	0.807 (0.787–0.828)	0.861 (0.833–0.889)	0.863 (0.81–0.92)	0.881 (0.858–0.905)	0.903 (0.858–0.95)	0.909 (0.871–0.948)	0.918 (0.893–0.943)	
X. Li et al 2017 [22]	Variables (Total: 167)	Prior TE	Ischaemic stroke	CHF	LV posterior wall thickness	LV ejection fraction	cholesterol	MI	ICH	Drug use for rate control	Years since last TIA	
G. Lip et al 2021 [51]	Variables (Total:?)	Multimorbid index	Age group	Chadsvasc	Age	CHads2	Medicare status	Prior history of stroke	Exposure time from last prior stroke	Exposure time AF	Multimorbid index2	
	Importance	1	0.872	0.865	0.856	0.72	0.572	0.511	0.508	0.345	0.242	
A. Papadopoulou et al 2024[45]	Variables (Total:107)	No. Hospital admissions	Age	AF_SCORE	Height	Weight	Waist circumference	Sex	No. Medications	Albumin	Aspirin	
	Importance	0.527	0.481	0.214	0.106	0.092	0.067	0.059	0.053	0.050	0.048	

HTN – hypertension; Hb – haemoglobin; TIA – transient ischaemic attack; eGFR – estimated glomerular filtration rate; IHD – ischaemic heart disease; UTI – urinary tract infection; DM – diabetes mellitus; PE – pulmonary embolism; CKD – chronic kidney disease; TE – thromboembolism; CHF – Congestive heart failure; LV – left ventricle; MI – myocardial infarction; ICH – intracerebral haemorrhage; AF – atrial fibrillation.

Table 2 lists the thirty-eight studies assessing the risk of stroke in the general population. The highest performing model in each study is listed in bold, along with its performance statistics. These included groups with and without diagnoses of AF. The area under curve (AUC) for test data varied from 0.64 to 0.99. Nine studies had an AUC < 0.80, 7 studies had an AUC between 0.80 and 0.90 and 12 studies an AUC > 0.90. Ten studies did not provide a value for the AUC but reported accuracies between 72 % and 99 %. Fourteen of 38 studies provided values for sensitivity and specificity. Values for sensitivity ranged from 66 % to 100 % (and one outlier of 6 % [48]). Values for specificity ranged from 33 % to 99 %. Four studies compared performance against clinical risk stratification tools.

Table 3 lists the eleven studies that assessed the risk of stroke in those with AF (Table 3). Again, the highest performing model in each study is listed in bold, along with its performance statistics. Nine of the eleven compared ML against clinical scoring methods such as CHA_2DS_2-VASc. All nine of these studies concluded that ML was superior to current clinical scoring tools in predicting the future risk of stroke. For studies assessing the risk of stroke in those with AF, the AUC varied between 0.63 and 0.89. Five studies had an AUC < 0.70 and five studies had an AUC > 0.70. One study stated that their model was able to accurately predict primary stroke but did not report metrics on model performance [49]. Three studies reported values for sensitivity and specificity [44,46,50]. The mean value for sensitivity was 63 %. The mean value for specificity was 69 %.

3.3. Model validation

Thirty-three of 49 studies performed cross validation. Generally, this was 10-fold cross validation. Studies typically split their datasets into training and validation/test subsets in a ratio of 3:1. Only four studies performed external validation of their models [23,36,44,51].

3.4. Balancing the dataset

Balancing the ratio of cases to controls is an essential step that prevents models from overtraining to the majority class [52]. 8 studies stated that they adopted synthetic-minority-oversampling-technique (SMOTE) to address unbalanced datasets [25–27,31,33,53–55]. This is a strategy in which an underrepresented group in the dataset is identified. New, artificial cases for this group are then created based on the characteristics of the existing members of the group. This enables the count for the underrepresented group to be inflated, and thus balancing the dataset [52]. Several studies also experimented with using random-over-sampling, random-under-sampling, assigning class-weights or novel balancing methods [31]. Generally, assigning class-weights and/or using SMOTE delivered the best performance.

3.5. Explainability and interpretability

Explainability analysis provides insight into how models arrive at their estimated risk scores. For the purposes of this review a study is considered interpretable if variables that contribute to model performance and the type of model used are reported [16]. Studies were considered explainable if they reported variable importance analysis and assessed the degree in which variables contribute towards model output [16]. Overall, 32 of 49 studies performed some form of explainability analysis and ranked variables in order of importance. Thirteen calculated importance values for variables. Forty-seven of 49 studies reported the variables used in model design and were considered interpretable. Two studies provided no information on variables used in model design.

3.6. Calibration

Calibration is the process of aligning the probabilities estimated by

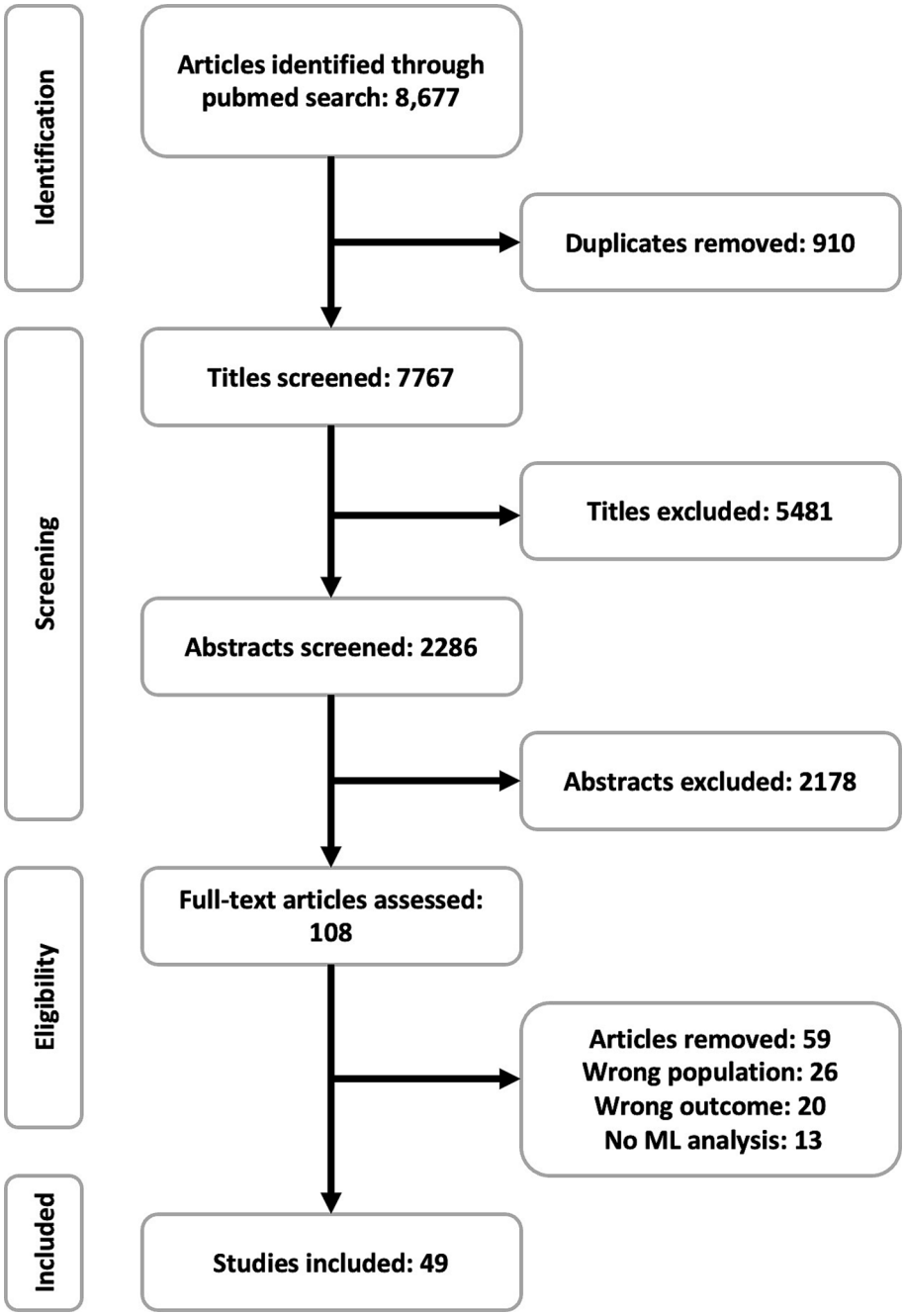


Fig. 1. PRISMA flow diagram.

models with real life events [56]. Twelve of 49 studies tested for good model calibration. Five studies tested for good calibration with graphical and statistical testing, such as Hosmer-Lemeshow test or Brier score. Six studies tested for good calibration with only graphical methods, such as Platt scaling. One study did not specify a method for testing for good calibration [43].

Ten of the Twelve studies that performed calibration reported that their models were well calibrated. Six of these only stated they were well calibrated in the text and provided no calibration metrics [41,43,51,57–59]. Four of these studies reported calibration statistics (chi-square value for Hosmer-Lemeshow statistic or brier score) but did not provide p-values, making interpretation difficult [23,37,38,42].

Two studies did not comment on the results of their calibration testing [43,60]. One study reported that their model was poorly calibrated [35].

3.7. Machine learning model performance

Overall, there was no clearly superior model, however GBDT tended to outperform other models, yielding greater AUCs and better model calibration. A total of 19 studies compared GBDT against other ML models. GBDT outperformed other ML models in ten of these studies. Deep neural networks (DNN) also yielded a higher AUC and outperformed other ML models in 5 of 8 cases [21,31,53,54,61].

On average, models were able to better predict the future risk of stroke in the general population, compared to patients with AF. The mean AUC for studies including patients in the general population was 0.85 (range 0.64–0.99) and 0.73 (range 0.63–0.89) for studies including only patients with AF.

3.8. Variable importance

Tables 4 and 5 summarise the important variables used in model construction for the general population and patients with AF, respectively. In both instances, the most important variables included age, gender, body mass index (BMI), hypertension, history of stroke/TIA, history of cardiovascular disease, smoking and alcohol use. Biochemical markers, such as glucose, lipid profile, CRP and creatinine were also deemed important in model performance. The addition of carotid ultrasound and physiological parameters (e.g., BMI, blood pressure value and heart rate) also appeared to be useful in improving model performance.

3.9. Novel biomarkers

Several new markers were also identified among the included studies. One optimal classification tree model showed that in patients with a history of cardiovascular disease, the presence of unspecified T-wave abnormalities on ECG significantly increased risk of stroke [23]. Two studies showed that augmentation of carotid ultrasonography with routine clinical data significantly improved model performance. One study using a GBDT saw that in patients with non-valvular AF, a slight improvement in overall AUC was seen when routine clinical data was augmented with cardiac ultrasound findings. Predictive variables included left-ventricular-wall-thickness, left ventricular mass, aortic regurgitation and mitral regurgitation [50]. Another study using a Sub-Population Optimization and Modelling Solutions (SOMS) tool identified that amongst patients with AF, urinary tract infection is a significant predictor of stroke (variable importance of 4.6 %) [49]. Addition of blood test data also appeared to marginally improve model performance. These included markers of hepatic and renal function (creatinine, urea, bilirubin, and albumin), inflammation (CRP, haematocrit, lymphocyte count), coagulation (platelet count and fibrinogen), glucose regulation (hba1c, fructosamine and glucose) cholesterol and haemoglobin [27,53,62–66].

4. Discussion

4.1. Study findings

This article aims to provide a systematic review of the efficacy of ML in predicting the future risk of stroke from routine hospital data. We reviewed 49 articles from May-2015 to December-2024. Eleven of these articles assessed the future risk of stroke in populations with AF, and 38 in the general population.

All studies endorsed the use of ML in predicting the future risk of stroke, including 13 articles that compared ML against clinical scoring tools, such as CHA_2DS_2-VASc.

These findings suggest that ML could be a useful tool in stratifying the risk of stroke in patients from routine hospital data, however further comparison with current clinical risk stratification tools is required. We observed that ML may be particularly useful in predicting the risk of stroke in the general population, since models appeared more accurate in predicting the risk of stroke in those with or without AF, compared to those with exclusively AF. Furthermore, no accurate scoring system to stratify the risk of stroke in those without AF currently exists [67]. Given only 20 % of ischaemic strokes are associated with AF and no accurate scoring system exists to stratify stroke risk for the general public, this makes a case for external validation of models that are able to stratify stroke risk in the general public or in those without AF [6].

It is unclear why ML models were able to predict stroke with a greater degree of accuracy in the general population, compared to patients with AF. However, this may be due to variable dominance, where a dominant predictive variable over-influences the model learning process, shadowing the input from other variables and reducing generalisability [68]. AF increases the risk of stroke by approximately 5 times and hence AF variable dominance could explain the poorer performance and lack of consistency seen in variable importance in these models [6].

It is difficult to comment on which models perform best in predicting stroke, however GBDT, DNN and RF tended to outperform other model types. It is instead more important for experimenters to explore their dataset and select models most suited to handling a particular data type. For example, GBDT and DT may be better suited to handling a combination of continuous and categorical variables with missing variables, whilst DNN may be better suited to handling complex image data [16]. Generally, studies adopted the sensible approach of trialling a variety of models on their datasets, comparing the performance of each one.

To better understand what variables may be useful in predicting stroke we reviewed features that demonstrated high importance values in AF and general population studies. Amongst the top ranking, included male gender, older age, smoking history, blood pressure (which generally showed a bimodal association with stroke), previous stroke/TIA and history of heart disease, physiological observations, and BMI. These variables have previously been identified as major risk factors for cardiovascular disease and stroke, and several are included in the CHA_2DS_2-VASc score and QRISK3 [3,10]. Although not specified, a number of studies were also able to improve their prediction accuracy by including variables from cardiac and carotid ultrasound [22,64]. Indeed, left ventricular hypertrophy, left atrial appendage thrombus and carotid stenosis are all strong risk factors for stroke [69].

ML also proved effective at identifying novel risk factors for stroke. The areas which showed most promise in novel risk factor identification included ECG and laboratory test data, which are discussed as follows [23,50,65].

Unique ECG patterns such as premature atrial and ventricular ectopics, prolonged QT interval and ECG derived left ventricular hypertrophy have been reported to be associated with stroke [70]. Ectopic beats and prolonged QT intervals are commonly associated with tachyarrhythmias which may lead to ischaemic stroke via similar akinetic mechanisms to AF or could be precursors to AF itself [70]. Electrophysiological left ventricular hypertrophy is a sequela of hypertension and an indicator of heart failure, which have both been identified as an independent risk factor for stroke. ECG T-wave inversion, ST depression and Q wave abnormalities have been associated with stroke risk. Given these are markers of ischaemic heart disease they may reflect a state of high cardiovascular risk and atherosclerotic burden [70]. Aortic regurgitation and mitral regurgitation were also all reported as risk factors for stroke [23,50]. These factors are also significant identifiers for cardiac structural abnormalities and risk factors for AF, suggesting a causal link with stroke [71,72].

Several biochemical markers including creatinine, CRP, glucose, and lipid profile were also identified as stroke risk factors [62,64,65]. Respectively, creatinine, glucose and lipid profiles are useful markers for kidney disease, diabetes, and hypercholesterolemia, all of which are major risk factors for stroke [73–75]. High CRP has been historically associated with cardiovascular disease and ischaemic stroke [64]. The mechanism underlying this remains elusive, but is likely due to the hypercoagulable states associated with Inflammation and endothelial dysfunction [76]. Similarly, Alanazi et al 2022 reported other markers of inflammation, such as leucocyte count, mean red cell width and haematocrit to be associated with ischaemic stroke [65]. Indeed, markers red-cell-count and haematocrit have previously been associated with ischaemic stroke as they reflect states of hyperviscosity, polycythaemia and/or dehydration, which are also associated with stroke [52,77,78].

It is reassuring that these risk factors have previously been thought to be associated with stroke and appear to relate to plausible clinical and pathophysiological explanations. This highlights that ML is likely capable of generating clinically relevant results that reflect true disease processes. Conversely the relative lack of novel risk factors identified is a concern and may imply that ML's ability to discriminate novel disease modifiers (or patterns of modifiers) may be limited.

4.2. Limitations in current ML research

In this review we identified several limitations to using ML to predict the future risk of stroke. Common pitfalls included poor generalisability, overfitting and a lack of calibration, validation and explainability analysis. These are discussed as follows.

Firstly, models often lacked generalisability. Many studies did not provide sufficient descriptive statistics of their cohort or study design. Information on data collection, primary outcome definitions, variables, observation period and exclusion criteria were often missing from studies. Studies often failed to delineate if their models aimed to predict "primary" stroke risk in patients with no history of stroke or TIA versus "secondary" stroke risk in patients with a history of stroke or TIA. This is an important consideration given a history of previous TIA/stroke is a significant risk factor for future stroke [79]. A large proportion of studies did also not provide values for sensitivity of specificity, which are essential markers required to assess feasibility for models in the context of public screening [80].

Secondly, models were often not calibrated. Calibration is an essential step in developing models where the output is a meaningful probability of an event occurring and can be compared against other models in the research and clinical field [56]. Although ML models appeared to outperform current clinical risk stratification tools, such as CHA_2DS_2-VASc, this is difficult to ascertain for sure. Only 12 of 49 assessed whether models were well-calibrated and 7 of these performed model calibration. Of note only 13 studies compared ML model performance against clinical risk stratification tools, hence it is difficult to comment on the true usefulness of ML in predicting future stroke risk in the clinical setting.

Thirdly, only four models performed external validation. This is an important consideration as many study cohorts were not representative of the world population. For example, a majority of the studies were based in Asian cohorts, highlighting the need for replication amongst other ethnic groups. Studies also often had observation periods less than five 5 years, raising the concern of censoring.

Fourthly, studies lacked explainability analysis. Explainability refers to the extent in which the inner workings and decisions of a model can be understood and interpreted [16]. It is essential in clinical research since it helps us identify pertinent risk factors that should be included in future stroke prediction studies and potential targets for future therapies. Without it, our ability to gain insight into stroke prediction and expand research in this area is limited [81]. Only 16 of 49 studies were either intrinsically explainable (i.e. LG) or performed explainability analysis such as Shapley Additive exPlanations (SHAP) [82].

Fifthly, there was a high suspicion of overfitting for many models since eleven studies reported AUC > 0.9. This may be due to small sample sizes [83]and inclusion of numerous variables [16], many of which were missing data and lacking key stroke-related variables (e.g., cholesterol, family history, smoking). Several studies performed feature selection analysis, but heavy weighting of closely related variables often led to multicollinearity issues. Most studies did not comment on regularization techniques to simplify their model [16] or the hyperparameters used to train their models, hence it is difficult to comment if these factors contributed to overfitting.

Finally, there is a lack of prospective validation of models and no RCTs which look to assess the roles of lifestyle and/or drug interventions in individuals ML models deem high risk of stroke. Given ML appears capable of accurately predicting stroke risk in both the general population and those with AF, we are hopeful that ML will play a key role in identifying those who will benefit from early treatment with drugs such as statins, antiplatelet and anticoagulants. These RCTs will also help identify the barriers of implementing ML into routine clinical care, such as job security, medical liability, effective communication with patients and sale of data/models to external organisations [84].

4.3. Study strengths and limitations

The main strength of this review is that we sought to only review higher level research investigating the role of ML in predicting the risk of stroke in adults from routine hospital data. We therefore included only original research articles with a PUBMED ID where routinely collected (e.g. demographics, ECG, echo, CT/MRI and blood tests) clinical data was used to predict stroke risk in adults. Therefore, studies without a PUBMED ID, study protocols, animal studies, participants < 18 years or studies that trained models solely on datasets from specialised tests (e.g. only carotid doppler data) were not included.

The main limitation is that the included articles were highly heterogeneous with respect to participant demographics, follow-up time and model design. This may explain the variation seen in results and makes true comparison of ML algorithms or identifying novel risk factors challenging. This should be considered when interpreting the findings of this review and further highlights the need for a standardised approach to ML model design within the healthcare setting.

Our search did not identify any articles that focused on the role of brain imaging in predicting future stroke risk. A further focused search for ML studies assessing the role of CT/MRI brain imaging in predicting future stroke risk also failed to identify any relevant articles. Given the significant risk associated with conditions that can be identified via imaging modalities, such as previous stroke, small vessel disease and cerebral amyloid angiopathy, CT/MRI brain imaging is likely to serve an important role in predicting stroke risk [85].

5. Conclusion

Based on the findings of our review, ML appears to be capable of accurately predict stroke from routine hospital data in both general and AF specific populations. Where assessed, ML also outperformed traditional clinical scoring systems, such as CHA_2DS_2-VASc, and hence could prove valuable in the clinical setting, but further validation is needed here. We also observed that ML was useful in identifying novel risk factors for stroke from blood test, ECG and ultrasonography data, which may help refine future risk stratification tools and aid new therapeutic discoveries.

Despite this, the role of ML in the clinical setting remains limited. This is largely due to methodological pitfalls in dataset quality, model calibration, external validation and explainability analysis which restricts the true validity and usefulness of these models in the clinical setting.

In order to address these issues and improve the quality of ML research in healthcare, future research should first focus on robust study design by adhering to the EQUATOR reporting standards and encourage interdisciplinary collaboration between clinicians and data scientists throughout the study design process. Next, external validation of models in the form of prospective RCTs is required, which will offer insight into the true utility of ML technology within the clinical setting.

In summary, while ML has already shown its potential in stroke prediction, substantial improvements in database quality, model explainability and validation of models in the clinical setting is required before integration of ML within routine healthcare. We therefore make the following recommendations for future study.

5.1. Future research

As outlined by EQUATOR guidelines, journals should consider adherence to the TRIPOD checklist for prediction models, prior to publication [47]. We also feel that there are several points not included in this checklist that articles should also outline, these include: 1) a description of feature engineering and feature selection techniques; 2) explainability analysis and discussion of important variables; and 3), methods to improve generalisation and prevent overfitting (e.g. regularisation). Collectively this guidance will ensure that future research is

of high quality and medically relevant. When developing ML models both clinicians and data scientists should also work in collaboration and peer review should be undertaken by individuals with appropriate expertise in ML.

Future research should also aim to integrate other investigations routinely performed in stroke. Our group are currently combining CT, MRI, ECG, echocardiography, and laboratory test data into a single ML model to predict future risk of stroke, in the hope this combination will improve the overall predictive power [86]. We also feel that future models should aim to externally validate their findings where possible. If external validation is not possible this should be highlighted as a limitation.

Analyses for secondary outcomes, such as bleeding risk associated with direct-oral-anticoagulant (DOAC) or antiplatelet therapy should also be performed. This may help to better inform clinical decision making and personalise stroke care. One effective strategy here might be the development of multilabel ML models capable of simultaneously predicting stroke and major bleeding risk [58].

Finally, prospective RCTs to evaluate the impact of lifestyle interventions and/or drug therapy in patients identified as high-risk by ML models is needed. These RCTs may be useful in evaluating the clinical utility of ML in stroke prevention, along with highlighting potential barriers of implementing ML into routine healthcare.

CRediT authorship contribution statement

William Heseltine-Carp: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Megan Courtman:** Writing – review & editing. **Daniel Browning:** Writing – review & editing. **Aishwarya Kasabe:** Writing – review & editing, Methodology, Formal analysis. **Michael Allen:** Writing – review & editing. **Adam Streeter:** Writing – review & editing. **Emmanuel Ifeachor:** Writing – review & editing. **Martin James:** Writing – review & editing. **Stephen Mullin:** Writing – review & editing, Supervision.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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